

Vanguard Civil Infrastructure

Document: Master Technical Specification & Site Compliance Protocol **Project Focus:** The Horizon Viaduct Initiative (Project Code: HV-2026) **Classification:** STRICTLY CONFIDENTIAL - INTERNAL ENGINEERING TEAM ONLY **Effective Date:** April 2026

1. Project Overview and Scope

This document outlines the strict engineering and safety protocols for Vanguard Civil Infrastructure, with a primary focus on the **Horizon Viaduct Initiative (HV-2026)**. The Horizon Viaduct is a 1.8-kilometer multi-lane suspension and beam bridge designed to cross the Silver Creek Valley.

- **Approved Budget:** \$425.5 Million USD.
- **Target Completion Date:** November 15, 2028.
- **Lead Architect:** Dr. Aris Thorne. Any deviation from the specifications listed in this document requires Tier-1 authorization from the Chief Structural Engineer.

2. Advanced Material Specifications

The structural integrity of Project HV-2026 relies on strict adherence to the following material grades:

- **High-Performance Concrete (HPC):** All bridge piers and submerged foundation structures must utilize **M55 grade** concrete with a maximum water-to-cement ratio of 0.35. A silica fume admixture (8% by weight of cement) is mandatory to prevent chloride penetration.
- **Deck Slab Concrete:** The upper driving surface will utilize **M40 grade** concrete, requiring a slump test result of 100mm to 120mm prior to pouring.
- **Steel Reinforcement:** Only **Epoxy-Coated TMT Fe550D** grade steel rebar is authorized for the main deck to resist de-icing salt corrosion.
- **Post-Tensioning Cables:** All internal tendons must be stressed to exactly 75% of their Ultimate Tensile Strength (UTS) a minimum of 21 days after the concrete pour.

3. Geotechnical and Foundation Protocols

Because the Horizon Viaduct crosses a valley with a high water table, foundation stability is the highest priority.

- **Pile Driving:** Friction piles must be driven to a minimum depth of **32 meters** below the riverbed. Piling operations must cease immediately if the standard penetration test (SPT) N-value exceeds 50 blows per 30cm, indicating bedrock.
- **Soil Compaction:** Embankment fill materials must be compacted to **95% of the Modified Proctor Maximum Dry Density**. Testing must occur every 500 cubic meters of fill placed.

4. Heavy Machinery and Crane Operations

- **Tower Crane Wind Tolerances:** Crane operations must be immediately suspended if sustained wind speeds exceed **35 km/h** (22 mph). If wind speeds exceed **60 km/h** (37 mph), the crane must be placed in "weather-vane" mode to freely rotate.
- **Electrical Clearances:** All mobile cranes, boom lifts, and excavators must maintain a strict minimum clearance of **10 meters (33 feet)** from overhead high-voltage power lines.
- **Emissions:** All heavy diesel machinery operating on-site must meet **EPA Tier 4 Final** emission standards. Equipment older than 2018 is banned from the site.

5. Environmental and Noise Regulations

Vanguard Infrastructure is committed to minimizing the impact on the local Silver Creek ecosystem.

- **Noise Limits:** Construction noise at the site boundary must not exceed **85 dB(A)** between the hours of 7:00 AM and 6:00 PM. Night work (6:00 PM to 7:00 AM) is strictly limited to **65 dB(A)**.
- **Silt Containment:** A dual-layer silt fence must be installed exactly 15 meters from the water's edge. Water turbidity (cloudiness) must be tested twice daily; if turbidity increases by more than 10% over the baseline, all earth-moving operations must stop.

6. Health, Safety, and Emergency Action Plan (EAP)

- **PPE Requirements:** High-visibility Class 3 vests, steel-toed boots (ANSI Z41 compliant), and Type 1 hard hats are mandatory.
- **Fall Protection:** Any worker operating at a height greater than **1.8 meters (6 feet)** must be tied off with a dual-lanyard full-body harness.
- **Code Red Evacuation:** In the event of a detected structural anomaly (e.g., early micro-cracking in a pier) or a localized seismic event, the "Code Red" alarm will sound. All personnel must evacuate to Assembly Point Bravo, located 500 meters north of the primary site office, within **4 minutes**.